Hariprashad Ravikumar

Machine Learning Research Engineer | PhD Physics, Dec 2026 | CUDA · HPC · LLM Systems
Las Cruces, NM · Available Jan 2027
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Experience

HAMR Modeling and Simulation Intern, Western Digital, San Jose, CA *(May 2026 – Aug 2026)*

- Shipped a full-stack physics simulation web app (Python, Dash, Plotly) that lets engineers compute signal-to-noise ratio using a physics/statistical model instead of on physical hardware; used daily in production by 40+ senior R&D engineers and subject matter experts across the company's US and Japan sites
- Cut per-experiment turnaround with real-time parameter sweeps, side-by-side run comparison, one-click PowerPoint/Excel export, and IndexedDB cross-tab persistence
- Packaged the tool as a PyPI-installable Python library, deployed it on Kubernetes through a Jenkins CI/CD pipeline, and integrated it into the internal engineering platform alongside related analysis tools
- Derived a closed-form model of magnetic grain switching from first principles, combining thermal activation and coherent rotation theory to predict switching time, signal noise, and error probability across repeated write cycles of a hard disk drive (HDD)
- Validated the model against Monte Carlo simulation, large-scale micromagnetic simulations, and experimental data; core findings are in preparation for peer-reviewed publication
- Identified causal drivers of storage performance through physics/statistical modeling, which let engineers predict and reduce data integrity errors computationally and removed a hardware test step from the R&D loop
- Presented results at PhD Expo 2026 (50+ attendees) and multiple internal engineering reviews (30+ attendees each), including a cross-site review with the company's Japan R&D team

Graduate Research Assistant, NM State University, Las Cruces, NM *(Aug 2021 – Present)*

PhD Project: Lattice Quantum Chromodynamics & Machine Learning Approaches

- Generated 30,000+ synthetic data points by solving Partial Differential Equations with Hybrid Monte Carlo simulations (Markov Chain stochastic process), then built an AI for Science pipeline that models the underlying physics with over 98% accuracy using symbolic regression
- Fit non-linear, high-dimensional models to those 30,000+ points with parallelized Bayesian inference across multi-CPU architectures, using automatic differentiation, full covariance matrices, and jackknife resampling for exact error propagation

Technical Projects

Transformer Pretrained From Scratch Hugging Face | GitHub

- Independently pretrained a 57M-parameter transformer from scratch in PyTorch, achieving 1.057 bits-per-byte (a tokenizer-agnostic metric, comparable across models with different vocabularies) versus 1.135 for a same-size Pythia-70M baseline, by building the full pipeline: custom tokenizer, sharded data loading, and fault-tolerant checkpointing on Google Cloud (GCP) Spot GPUs
- Deployed the trained model to Hugging Face Hub, eliminating recurring hosting costs by serving inference entirely client-side via Open Neural Network Exchange (ONNX) Runtime Web

Cost-Aware Multi-Agent LLM Router GitHub | Live Demo

- Built a calibrated LLM routing system, cutting inference cost 93.9% versus always calling the frontier model with no statistically distinguishable accuracy loss on a 50-prompt disjoint holdout, by training a logistic regression over logprob uncertainty, self-consistency dispersion, and MiniLM embedding-distance features, and validating it for honesty using Expected Calibration Error and Brier score
- Extended the system into a multi-agent conversational mode using LangGraph (planner, parallel specialist dispatch, synthesizer) with session and cross-session memory via Redis and pgvector, deployed as a FastAPI service with Postgres logging and an async circuit breaker on Google Cloud Run

Research Collaborations

Los Alamos National Laboratory - Lattice Quantum Chromodynamics *(May 2024 – Present)*

- Ported multi-terabyte Hamiltonian Monte Carlo calculations to parallelized C++/CUDA kernels on GPU-accelerated HPC clusters (NERSC Perlmutter) to cut per-job runtime
- Ran 75,000+ CPU/GPU compute hours through custom SLURM workflows built for job orchestration at that scale

North Carolina State University - Mathematical Physics *(Dec 2020 – Present)*

- Analyzed algebraic symmetry constraints with Mathematica symbolic computation on HPC clusters, published in Physical Review D

Selected Publications

Google Scholar

1. C.-R. Ji and **H. Ravikumar**, *"Interpolating conformal algebra in (1+1) dimensions,"* **Physical Review D** 113, 096018 (May 2026). DOI: 10.1103/prlq-4j1l

Technical Skills

Methods & HPC: Parallel Computing (GPU, MPI), Numerical Methods (PDEs, Monte Carlo, Regression)
Programming: Python, C++, CUDA, Bash, SQL
ML & APIs: PyTorch, TensorFlow, Scikit-learn, Pandas, Dash, Plotly, Flask, FastAPI, Numba
Cloud & MLOps: Azure, AWS (S3, Lambda), Hugging Face, Docker, Kubernetes, Jenkins, CI/CD, Git, SLURM

Education

PhD in Physics, New Mexico State University, USA *Aug 2021 – Dec 2026 (expected)*
MS in Physics, New Mexico State University, USA *Aug 2021 – May 2024*

Awards

- **2025 NMC Collaboration Grant**, awarded by the New Mexico Consortium for an independent research project in collaboration with scientists at Los Alamos National Laboratory
- **2023 George and Barbara Goedecke Physics Excellence Fund Scholarship**, awarded by the NMSU Physics Department

Leadership

- **Vice President**, Physics Graduate Student Organization (NMSU) *Aug 2025 – Sep 2026*
  Organized professional development events and served as primary liaison between 40+ graduate students and faculty

Relevant Graduate Coursework

- Quantum Computing, Advanced Computational Physics, Statistical Mechanics, Electromagnetic Theory I & II